

From Inverted Indices to Agentic RAG: How Retrieval Grew Up

Follow information retrieval from keyword matching to systems that decide for themselves what to look at.

For engineers building search or RAG systems who want to know which stage of the evolution their system is stuck at · 26 July 2026

The one- or two-page version: what matters, the topics it covers, and every term defined. Read the full lesson for the explanations and worked examples.

[Watch the source video](#)[Read the full lesson](#)[Read the condensed version](#)

What matters

- ✓ An inverted index maps terms to the documents containing them, which is what makes keyword search fast.
- ✓ BM25 ranks by term frequency, inverse document frequency and document length, and remains a strong baseline decades on.
- ✓ Lexical search treats words as symbols, so synonyms, ambiguity and intent are invisible to it.
- ✓ Embeddings are learned from context, which is why 'espresso' lands near 'coffee' and far from 'house'.
- ✓ Hybrid retrieval combines lexical precision with semantic recall because each covers the other's failures.
- ✓ LLMs predict likely tokens rather than retrieving facts, so their knowledge is frozen and unattributable.
- ✓ Naïve RAG is linear — embed offline, retrieve once, generate — and is only ever as good as that single retrieval.
- ✓ Agentic RAG makes retrieval a tool invoked during reasoning, so the system can iterate, verify and stop when it has enough.

What it covers

01 Keyword search: inverted indices and ranking 0:00

The earliest systems answered one question — where does this word appear — using a mapping from terms to documents, then ranked the hits by statistical importance.

02 Where lexical matching breaks 1:24

It treats words as symbols, not as meaning. Synonyms, ambiguity and intent are invisible.

03 Semantic search: meaning as geometry 2:09

Represent text as vectors in a high-dimensional space where similar concepts sit close together, and relevance becomes distance.

04 Hybrid retrieval: precision and recall together 2:49

Semantic search did not replace keyword search. Systems combining both emerged because each covers the other's failures exactly.

05 LLMs predict — they do not retrieve 3:31

A language model returns the most likely continuation based on learned patterns. It has no index, no sources and no notion of looking something up.

06 Naïve RAG: the linear pipeline 4:20

Search for relevant documents, augment the prompt with them, generate the answer. Simple, effective, and only as good as one retrieval.

07 **Advanced RAG: rerankers, query rewriting, hybrid** 5:44

Within a short period the simple concept accumulated real machinery — but the pipeline stayed fixed.

08 **Agentic RAG: retrieval as a tool inside reasoning** 6:29

Retrieval stops being a fixed step before the answer and becomes something the system invokes, repeats and stops as part of thinking.

Glossary

Inverted index — A mapping from each term to the documents containing it — the reverse of the natural document-to-terms direction, which is what makes lookup fast.

BM25 — The standard lexical ranking function, refining TF-IDF with saturating term frequency and document-length normalisation.

Semantic search — Retrieval by proximity between embeddings, so documents match on meaning even when they share no words.

Hybrid retrieval — Running lexical and semantic retrieval together and fusing the results, since each covers the other's characteristic failures.

Bi-encoder — An embedding model that encodes query and document separately, allowing the corpus to be indexed in advance.

Reranker — A second-stage scorer, usually a cross-encoder, applied to a retrieved shortlist to reorder it more accurately.

Naive RAG — The original linear pipeline: retrieve once, augment the prompt, generate. No feedback and no second attempt.

Multi-hop question — A question whose answer requires a second lookup that depends on the result of the first. Beyond the reach of single-shot retrieval.

TF-IDF — Term frequency times inverse document frequency: a term matters more if it is frequent in this document and rare across the corpus.

Lexical search — Retrieval by literal term matching. Precise on exact strings, blind to synonyms and intent.

Embedding — A high-dimensional vector representing text, learned so that items appearing in similar contexts occupy similar positions.

Reciprocal Rank Fusion (RRF) — Combining several ranked lists by summing $1/(k + \text{rank})$ per document, avoiding the need to make incompatible scores comparable.

Cross-encoder — A model that reads query and document together and scores the pair. More accurate than a bi-encoder and too slow to run over a full corpus.

Query rewriting — Reformulating a short or conversational question into a fuller standalone query before retrieval, to improve recall.

Agentic RAG — Retrieval invoked as a tool during reasoning, so the system decides whether, where and how often to search, and when it has enough.

Explanations are AI-generated. Check anything you intend to rely on.